



Voronoi average number of vertices ≤ 6

Asked 12 years, 6 months ago Modified 12 years, 5 months ago Viewed 3k times



My text says "the average number of vertices of the Voronoi cells is less than six". Then it creates the vertex "at infinity", connects the half-infinite edges to this vertex and shows the equation:

3

$$(v + 1) - e + n = 2$$



where v = number of vertices (before creation of the one at infinity), e is the number of edges, and n is the number of point sites. It then shows the inequality:



$$2e \geq 3(v + 1)$$

I've probably been staring at this too long, but how to show the average number of vertices of the Voronoi cells is less than six?

computational-geometry

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edited Mar 1, 2012 at 17:21



Michael Hardy

1

asked Mar 1, 2012 at 15:42



PeteUK

1,560 1 15 27

1 Answer

Sorted by: Highest score (default)



Let F_k be the number of Voronoi cells with k vertices. By counting edges you get $2e = 3F_3 + 4F_4 + 5F_5 + \dots$. The average you seek is

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$$\frac{3F_3 + 4F_4 + 5F_5 + \dots}{F_3 + F_4 + F_5 + \dots} = \frac{2e}{n} \leq \frac{2(3n - 6)}{n} = 6 - \frac{12}{n} < 6$$



Here I've used that $e \leq 3n - 6$, which follows from



$$2(v + n - 1) = 2e \geq 3(v + 1) \implies v \leq 2n - 5 \implies e = v + n - 1 \leq 3n - 6$$



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edited Mar 1, 2012 at 16:37

answered Mar 1, 2012 at 16:31



lhf

218k 19 246 568

5 Easier: Let $\deg(f)$ be the number of edges of face f . Then $\sum_f \deg(f) = 2e \leq 6n - 12$ by Euler's formula. The pigeonhole principle now implies that at least one face f has $\deg(f) < 6$. – JeffE Mar 5, 2012 at 21:29